Frontiers of Science

ML1 Activity
Making, Folding, and Mutating Proteins: An Alzheimer's Case Study

This activity focuses on the following learning objectives:

1. Develop a working vocabulary of key terms in (1) Biophysical Chemistry: chemical bonds (covalent, ionic, and hydrogen bonds), macromolecules, biomolecules, monomers (amino acids and nucleotides), biopolymers (DNA, RNA, and proteins), protein structure (primary; secondary: alpha helix and beta sheet; and tertiary), and nucleic acid structure (primary; secondary: double helix; and tertiary)
2. List the major biomolecules involved in each of the following molecular processes in the cell: replication, transcription, and translation. Discuss the roles of these processes and biomolecules in the context of the Central Dogma of Molecular Biology.
3. Describe how scientists use computational tools, such as PyMOL and AlphaFold, to access publicly available databases in efforts to understand diseases and how to combat them.

Part 1: Central Dogma

Complete the table below for replication, transcription, and translation.

Process	Replication	Transcription	Translation
Molecular Machine	DNA polymerase	RNA polymerase	ribosome
Molecular machines are composed of…	proteins	proteins	rRNA + proteins
Input Biopolymer (= template)	DNA	DNA	mRNA
Output Biopolymer (= result of process)	DNA	RNA (mRNA)	polypeptide (protein)
Conversion rule	DNA -> DNA via complementary base pairing (A–T, C–G)	DNA -> RNA via base pairing (A–U, T–A, C–G, G–C)	mRNA codons (3 bases) -> amino acids (genetic code)
Monomers of Output	deoxyribonucleotides (dATP, dTTP, dCTP, dGTP)	ribonucleotides (ATP, UTP, CTP, GTP)	amino acids

Now answer questions A-D below:

Alzheimer's disease (AD) is one of the most common types of dementia, accounting for an estimated 60 - 80% of cases. AD is caused by the accumulation of two proteins: **beta-amyloid** and **tau**. When these proteins accumulate outside of neurons, they form large structures called fibrils, which damage synapses and ultimately neurons themselves, leading to symptoms of memory loss, issues with language, and other cognitive impairments.

Below is the DNA sequence for the **amyloid beta protein AB40** that accumulates in AD. As with many proteins involved in diseases, the AB40 protein plays a role in healthy neuron function. In healthy neurons, AB40 helps facilitate synaptic plasticity in memory formation. Let's use the three processes in the Central Dogma to make AB40!

Below is a portion of the DNA sequence of the AB40 protein:

CTA CGG CTT AAG TCC GTG CTG TCA CCG ATG CTC CAG

A. *Bang!* You're a DNA polymerase! Make another copy of the piece of AB40 DNA and write down here:

   GAT GCC GAA TTC AGG CAC GAC AGT GGC TAC GAG GTC

B. *Poof!* You're an RNA polymerase! Transcribe the original piece of DNA into a messenger RNA:

   GAU GCC GAA UUC AGG CAC GAC AGU GGC UAC GAG GUC

C. *Zap!* you're a ribosome! Translate this messenger RNA into a protein, using the mRNA codon table below. To make the sequence easier to use later in the activity when putting it into AlphaFold, record the amino acids using the one letter codes (shown in purple in the chart below)

   GAU GCC GAA UUC AGG CAC GAC AGU GGC UAC GAG GUC
   D A E F R H D S G Y E V

   peptide here: DAEFRHDSGYEV

An mRNA codon chart (also known as a genetic code table) is a visual representation that shows the correspondence between mRNA codons (three-nucleotide sequences) and the amino acids they code for. To use an mRNA codon chart, identify the first two bases of the mRNA codon on the chart's left side (First position) and top (Second Position), then use the third base to find the specific amino acid coded by that codon. For amino acids that can be encoded by multiple codons, the codons are grouped together with a bracket. Besides the three letter code for each amino acid, there are also one letter codes. This is practical especially for long chains of amino acids. The one letter code is written in purple on top of the three letter code, and you should use these one letter codes for your sequence.

Second Position

First Position	U	C	A	G	Third Position
U	UUU F Phe	UCU S Ser	UAU Y Tyr	UGU C Cys	U
U	UUC	UCC	UAC	UGC	C
U	UUA L Leu	UCA	UAA STOP	UGA STOP	A
U	UUG	UCG	UAG STOP	UGG Trp W	G
C	CUU L Leu	CCU P Pro	CAU H His	CGU R Arg	U
C	CUC	CCC	CAC	CGC	C
C	CUA	CCA	CAA Q Gln	CGA	A
C	CUG	CCG	CAG	CGG	G
A	AUU I Ile	ACU T Thr	AAU N Asn	AGU S Ser	U
A	AUC	ACC	AAC	AGC	C
A	AUA	ACA	AAA K Lys	AGA R Arg	A
A	AUG M Met	ACG	AAG	AGG	G
G	GUU V Val	GCU A Ala	GAU D Asp	GGU G Gly	U
G	GUC	GCC	GAC	GGC	C
G	GUA	GCA	GAA E Glu	GGA	A
G	GUG	GCG	GAG	GGG	G

D. For the rest of the activity, you'll need to include the rest of the sequence of AB40! Put the sequence you translated in the blank below to obtain the **full sequence** of the AB40 protein:

DAEFRHDSGYEV HHQKLVFFAEDVGSNKGAIIGLMVGGVV

Part 2: Protein Folding

Once proteins like AB40 are translated, they must fold into the correct structure to function properly. Many biophysical chemistry research groups study how AB40 folds and functions to understand how it switches from regulating synaptic plasticity in healthy neurons to blocking synapses in AD. One key aspect of this switch is the number of copies of the protein: in AD, multiple copies of AB40 accumulate and fold into different structures called fibrils. Let's apply our understanding of Molecules and Life and AlphaFold to explore this fascinating aspect of protein structure and function!

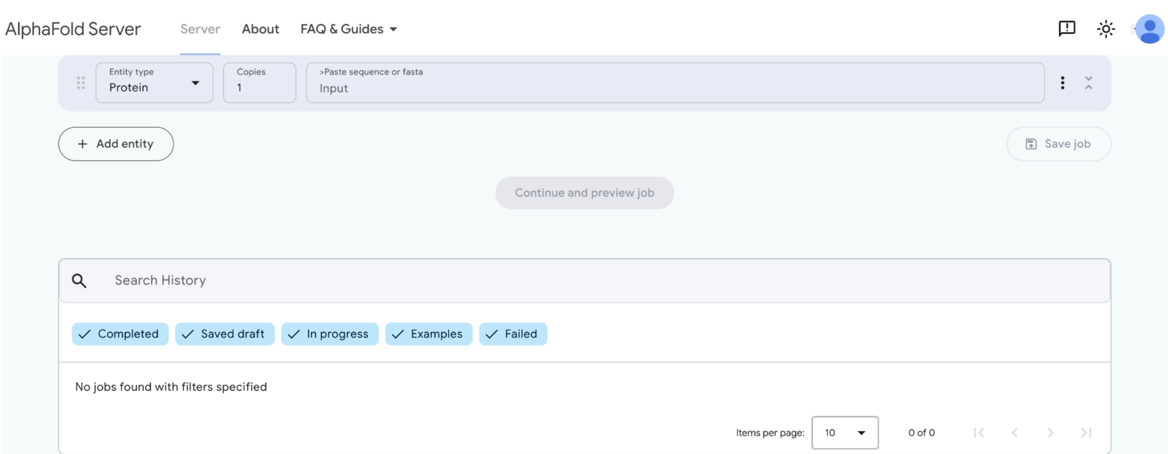
Answer the following questions below:

A. As we discussed in lecture, AlphaFold uses artificial intelligence to predict the structure of proteins based on their sequence. ***What kind of data is used to train the models in AlphaFold?***

AlphaFold is trained on big collections of protein amino acid sequences

B. AlphaFold is developed and owned by Google, and anyone can use it! We'll use AlphaFold to predict the structure of just one copy of AB40 first. Use this link to access it: https://alphafoldserver.com/

Click on the button that says "Continue with Google" and use your LionMail information to log in. On the following screen, you can enter your affiliation as "Columbia University" and accept the Terms of Service. You should see a page that looks like this:



Copy your translated protein sequence (it should be 40 amino acids!) into the box that says "Input". You can keep the entry type as "Protein" and Copies as 1. Once you're ready, you can click "Continue and Preview Job". For "Job Name", enter "**AB40_singlecopy**".

It will take a few minutes to generate a predicted structure. Once your job finishes, view the results by clicking on the name of your job. ***Paste a screenshot of your predicted structure below:***

1. ***What kind of structure does it make? What kinds of bonds hold this structure together?***

This single copy folds into a compact shape with short beta strands and flexible loops, it's not a long straight fibril. The structure is stabilized by Hydrogen bonds along the backbone

2. ***How confident is the prediction? What do you think this depends on? What would make this confidence go up?***

AlphaFold is usually more sure about stiff, well ordered parts and less sure about loose end regions, its confidence depends on how many similar proteins and known structures it can compare to, and it would go up if we had more good structures and more related sequences for AB40.

3. ***Why might it be helpful to researchers to know what this structure looks like?*** Remember: you didn't have to do ANY experiments in a lab to see this structure!

The single copy structure helps scientists see which parts of AB40 are on the surface or buried, it points to likely spots where other molecules or drugs might bind, and it guides which lab tests or mutations they should try next.

C. In Alzheimer's disease, multiple copies of AB40 accumulate outside neurons, forming different structures (fibrils, shown in Figure 1) that interfere with the synapse. These structures can also disrupt nearby neurons, leading to cell death. Let's explore what happens to the structure when there are multiple copies of AB40 around.

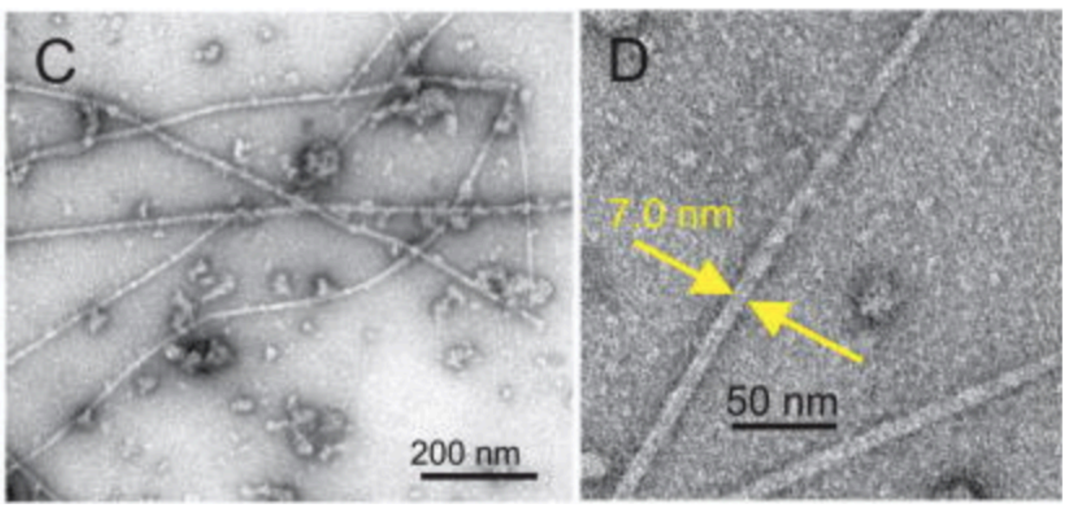


Figure 1. *Images of fibrils grown in a lab using fibrils extracted from the brain tissue of an AD patient.*

Return to the home screen with your first job, the sequence should still be in the "Input" box. Increase the number of copies to 10. For "Job Name", enter "**AB40_tencopies**".

Paste a screenshot of your predicted structure below:

1. ***What kind of structure does it make? Is it similar to the structure with one copy of the protein?***

With ten copies, AB40 is predicted to stack into a long, ordered bundle that looks more like an amyloid fiber, and this shape is more stretched out and regular than the single, compact chain.

2. Orient your protein structure so the individual copies of the protein are stacked on top of each other. Zoom in on the structure and hover over an amino acid until it is highlighted in red. Click on it to see the individual atoms and bonds that amino acid makes. ***What kinds of bonds hold the structure together? Include a screenshot with your answer.***

The many copies stick together through hydrogen bonds between nearby chains, through tight packing of greasy side chains in the middle, and through extra charge based and weak close contact forces at the joining surfaces.

3. ***How confident is this prediction?*** Compare it to the confidence level of the prediction for one copy. ***What do you think this tells us about the limitations of using AlphaFold?***

The ten copy model usually has lower and more uneven confidence than the single copy one, which means AlphaFold is less trustworthy for big, flexible clumps

Part 3: Mutations, mutations, mutations!

A small subset of Alzheimer's cases are familial Alzheimer's disease, an inherited form of the condition. Inherited diseases result from DNA mutations passed down to the next generation. These cases are marked by mutations in a single amino acid of the amyloid protein and often lead to early-onset symptoms. Let's use AlphaFold to explore how one of these mutations affects the fibril structure.

In one type of familial Alzheimer’s the 23rd amino acid (aspartic acid, D) is mutated to asparagine (N). ***Knowing that, answer the following questions below:***

1. ***What codon(s) encode aspartate (D)? What codons encode asparagine (N)?***

Aspartic acid (D) is coded by the mRNA codons GAU and GAC, asparagine (N) is coded by the mRNA codons AAU and AAC.

2. ***What mutation in the DNA sequence could cause this mutation in the protein?***

One example DNA change is to switch the coding strand codon from GAC to AAC, this single base change changes the amino acid from D to N at position 23.

3. ***Copy your original sequence below, find the 23rd amino acid, then insert and highlight the mutation below:***

   DAEFRHDSGYEVHHQKLVFFAENVGSNKGAIIGLMVGGVV

   Enter your mutated sequence into AlphaFold and the number of copies at 10. For "Job Name", enter "**AB40_tencopies_mutation**".

   Paste a screenshot of your structure below (it might help to explore and show multiple orientations of the protein):

4. ***What kind of structure do multiple copies of the mutated protein make? Is it similar to the structure of one copy? Or multiple copies of the unmutated protein?*** In your answer, be sure to mention specific features of the structure(s) that are similar or different.

The mutant protein with N at position 23 still forms stacked fibril like bundles, but the local packing and contacts around position 23 shift compared with the original D version.

5. ***What does this structure tell you about how individual amino acids affect the structure of proteins?***

These results suggest that even one amino acid change can alter charge or polarity, this can change how the chain folds or sticks to its neighbors, and it can affect how stable or harmful the fibrils are.

Part 4: Reflections

1. ***Why would having the structure of a fibril associated with Alzheimer's disease be helpful to scientists?***

Having the fibril structure helps scientists see how AB40 chains stack and stick together, find spots where drugs or antibodies might bind, and compare healthy and disease shapes to think of better treatments.

2. ***What kinds of experiments would you want to do to follow up on these predictions?***

To follow up, I'd grow these fibrils in test tubes and watch how fast they form, test how harmful they are to brain cells in a dish, and use tools like microscopes to see if the real shapes match the AlphaFold predictions

References

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2. Parihar, M. S.; Brewer, G. J. Amyloid Beta as a Modulator of Synaptic Plasticity. *J. Alzheimers. Dis.* 2010, 22 (3), 741-763.
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5. Adapted from: https://www.dawsoncollege.qc.ca/ai/portfolios/protein-folding-an-a-i-teaching-module/
6. Lu, J.; Qiang, W.; Yau, W.; Schwieters, C. D.; Meredith, S. C.; Tycko, R. Molecular structure of beta-amyloid fibrils in Alzheimer’s disease brain tissue. *Cell.* **2013**, 154 (6).